Week 37: Mirror twins

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

This view repeats, by problem, the answers of the print guide, week-37-facilitator.pdf, whose source is editable/facilitator-src/guide.md; its page references are to that guide. The overview, the physical gate, preparation, launch, proofs and hints are there. Change both together; check_guide_copy.py compares them.

Problems and routes

Problem	K–1	2–3	4–5	route
Hide a turn and match it again	P1	P1	P1	first
The mirror pair	P2	P2	P2	first
Which twin does a new model match?		P3		first
The view from A			P3	return
Make two letters equal	P3	P4	P4	K-1 first; 2-3 return; 4-5 return
A, A, B and C on each model	P4	P5		return
One pair that matches and one that cannot	P5			return
Change one A to D		P6		return
How many rotations?			P5	return
How many different labelled models?			P6	return

First route. K–1 P1–P3; Grades 2–3 P1–P3; Grades 4–5 P1–P2. **Return visits.** K–1 P4–P5; Grades 2–3 P4–P6; Grades 4–5 P3–P6.

Key, one entry per problem

Hide a turn and match it again

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. Two labelled copies of a rigid tetrahedron can always be matched by undoing the hidden turn, and matching means every corner agrees, not only the top.

K–1: The two models are true copies and always match after undoing the hidden turn. If they do not, inspect the labeling and rigidity. Having the same top label is insufficient; all other corners must also agree.

Grades 2–3: True copies always match; undoing the hidden rigid motion is one matching motion.

Grades 4–5: All four corners must agree. Once A is aligned, the cyclic placement of the other three labels matters, not merely which labels are present or which label is at the top.

The mirror pair

K–1 P2 · Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

Idea. A tetrahedron with four different corner labels cannot be turned to match its mirror image; with A on A, checking the three turns about A is a complete search.

K–1: The printed A B C D versus A C B D pair cannot match under rigid translations and rotations. It swaps B and C while keeping A and D in place in the construction. Accept a complete check: A must go on A, and all three turns about A fail. A young child is not expected to produce it; the adult can demonstrate the stable reversed order after exploration.

Grades 2–3: The mirror pair cannot match. “No turn has worked yet” does not by itself establish “no turn can work.” Accept a complete check: A must go on A, and all three turns about A fail. The A-nearest-eye cyclic order is a second reason, covering every allowed turn.

Grades 4–5: No rigid matching motion exists for the four-distinct-label mirror pair. Page 5 gives both a geometric and an algebraic obstruction.

Which twin does a new model match?

Grades 2–3 P3 · first route

Idea. The 24 ways to place four different labels fall into two classes and the mirror pair has one model in each, so every new labelling matches exactly one of the pair.

The new model matches exactly one of the two, so no model matches neither: four distinct labels make two classes, and the Problem 2 pair has one model in each. To check, put the new model’s A on each A and try the three turns.

The view from A

Grades 4–5 P3 · return route

Idea. Looking from A toward the opposite face, a turn keeps the cyclic order of B, C and D while a mirror reverses it, so the order is a certificate that no turn matches the mirror pair.

In the printed view from A, the original opposite face reads B, C, D counterclockwise, up to choice of starting place. The mirror reads B, D, C counterclockwise. A planar rotation of the view only changes the starting place, not this orientation. Use A nearest the eye on both; viewing one from the opposite side reverses the comparison and can create a false match.

Make two letters equal

K–1 P3 · Grades 2–3 P4 · Grades 4–5 P4 · K-1 first; 2-3 return; 4-5 return route · wording differs by band

Idea. Making any two of the four labels equal lets the mirror pair match, because the reflection that swaps the two equal corners combines with the mirror to give a turn.

K–1: Yes. After D becomes C, the pair is A B C C versus A C B C in the same positional order. A one-third turn around the A axis sends the first model’s front-left B to the second’s front-right B, while the two Cs occupy the two C positions. Choose the turning direction that aligns B; the other labels then agree.

Grades 2–3: D becoming C allows a match, as above. Every other pairwise merge also allows a match: AB, AC, AD, BC, BD and CD. Start each test from the original A B C D / A C B D mirror pair; successive cumulative merges test a different question. The name kept for the merged pair does not change the result.

Grades 4–5: All six unordered choices of two labels allow a rotational match after merging, with the original mirror pair restored each time. One proof uses the reflection swapping the two equal vertices, then composes it with the mirror comparison to obtain an orientation-preserving symmetry. Details follow on page 5.

A, A, B and C on each model

K–1 P4 · Grades 2–3 P5 · return route · wording differs by band

Idea. Every way of placing A, A, B and C on the tetrahedron matches every other by a turn, so no mirror pair can be made with a repeated label.

K–1: No impossible pair exists with exactly A, A, B, C on congruent regular models. Every assignment belongs to one rotational class. Children can demonstrate examples; page 5 supplies the all-cases reason.

Grades 2–3: All A A B C assignments match by rotation, so a partner cannot make a nonmatching pair under the stated model rules.

One pair that matches and one that cannot

K–1 P5 · return route

Idea. With four different labels, a copy matches and the mirror pair does not, so both outcomes can be built on purpose.

A matching pair can use identical labelings, with one model subsequently rotated. An unmatching pair is the four-label mirror pair from Problem 2. Check all corners, including the back.

Change one A to D

Grades 2–3 P6 · return route

Idea. Restoring a fourth label to a matching repeated-label pair can produce either class: matching corners give a match, mismatched corners give a mirror pair.

Both outcomes are possible. In the printed pair, the first model has A at top and front left; the second has A at top and front right. Change the top A on both to D: the new pair does not match. Change the top A on the first and the front-right A on the second: it matches. Changing both side As gives a nonmatch; changing first-side and second-top gives a match.

How many rotations?

Grades 4–5 P5 · return route

Idea. A regular tetrahedron has exactly 12 rotations: four places for one corner, then three placements of the rest.

12 rotations as vertex maps, including leaving the model still. Only the final mapping counts, regardless of the motion history.

How many different labelled models?

Grades 4–5 P6 · return route

Idea. The 24 labelings with four different letters form two classes of 12 under rotation, the two mirror forms.

Two different four-label models up to rotation. All 24 distinct assignments split into two classes of size 12. They are the two mirror forms; reflection is not allowed when deciding whether two assignments count as the same.